Sri Eshwar College of Engineering
Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

Department of Electronics and Communication Engineering

Course name : Project with Design Thinking /Course code: U23EC651/ Innovative multidisciplinary Project, Project with Design Thinking

Title of the project : Room Occupancy Monitoring

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Class / Section / Semester	:	**II ECE / C / IV**
Batch	:	**2024-2028**
Project Review	:	**First Review**
Date	:	**3.03.2026**
Project Guide with Designation	:	**Dr.K.N.Vijeyakumar, Professor/ECE**
Name of the Industry (if applicable)	:	**NIL**
Team Members	:	**Laksana A** **Nithishwaran S** **Pravin A** **Roobak S**

CHALLENGE STATEMENT

Lack of Real-Time Accuracy
Existing systems fail to accurately track people during fast, continuous entry and exit, leading to incorrect occupancy data.

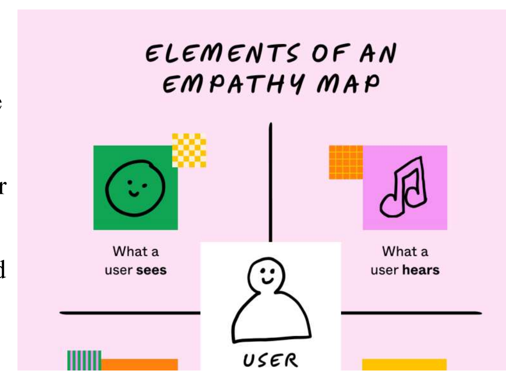
Processing Delays in Conventional Systems
Sequential execution in microcontroller-based solutions introduces latency, making them unsuitable for safety-critical environments.

High Power Consumption & Inefficiency
Camera-based and software-driven systems consume excessive power, limiting scalability for large buildings and continuous operation.

Privacy & Ethical Concerns
Vision-based monitoring systems compromise user privacy due to image capture and data storage, restricting real-world adoption.

EMPATHY MAPPING

- **Users**: Facility Managers, Event Coordinators, Administrators
- **See**: Overcrowding and lack of real-time occupancy visibility
- **Hear**: Safety regulations and demand for automated monitoring
- **Think & Feel**: Concern about safety and frustration with manual counting
- **Need**: Accurate, real-time, low-power, and privacy-safe solution



CONCEPT / SCOPE OF SOLUTION

- FPGA-based hardware system designed using Verilog HDL for real-time operation
- Two IR sensors used to detect entry and exit of people
- Finite State Machine (FSM) to identify movement direction accurately
- Real-time occupancy counter to update count (increment/decrement) instantly
- Capacity monitoring with alerts to prevent overcrowding and improve safety

LITERATURE SURVEY / BACKGROUND STUDY

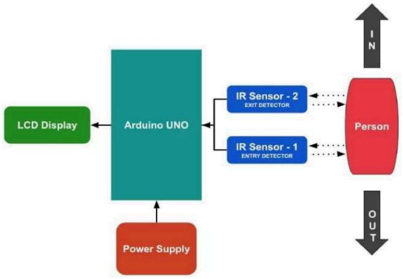
S. No	TITLE	AUTHOR	NAME OF ORGANIZATION & YEAR	DESCRIPTION
1	Infrared-Based Bidirectional Visitor Counter System	S. M. Metev, V. P. Veiko	IEEE / Embedded Systems, 2018	• Uses IR sensors to detect entry and exit based on beam interruption • Implemented using microcontrollers with sequential processing. • Limited accuracy during fast movement.
2	Arduino-Based Smart Room Occupancy System	A. Kumar, R. Singh	International Journal of Engineering Research, 2019	• Low-cost system using Arduino and IR sensors • Easy implementation and suitable for small applications. • Suffers from processing delays and timing issues.
3	IoT-Based Occupancy Monitoring and Energy Control System	M. S. Hossain, G. Muhammad	IEEE IoT Conference, 2020	• Integrates occupancy detection with IoT for remote monitoring. • Controls lighting and ventilation automatically. • Depends on network connectivity and increases system complexity.
4	8051 Microcontroller-Based Visitor Counter	R. S. Gaonkar	Embedded Systems Journal, 2017	• Uses basic IR sensors with 8051 controller • Simple design for classrooms and offices. • Not suitable for real-time high-speed movement detection.
5	Computer Vision-Based People Counting System	Z. Zhang, Y. Wang	IEEE Transactions / Computer Vision, 2021	• Uses cameras and deep learning for high accuracy. • Works well in crowded environments • High power consumption, cost, and privacy concerns.
6	Smart Building Occupancy Detection System	J. A. Stankovic, I. Lee	Smart Cities Journal, 2022	• Focuses on energy-efficient building management. • Uses multiple sensors for occupancy tracking • Requires complex integration and higher system cost.

EXISTING METHOD

Microcontroller-Based IR Counter (Arduino / 8051)
Uses IR sensors + microcontroller to count people entering/exiting

Camera-Based People Counting (Computer Vision)
Uses cameras + AI (YOLO / Deep Learning) to detect and count people

PIR Sensor-Based Occupancy Systems
Uses Passive Infrared (PIR) sensors to detect human presence and motion



PROPOSED / INNOVATIVE METHOD

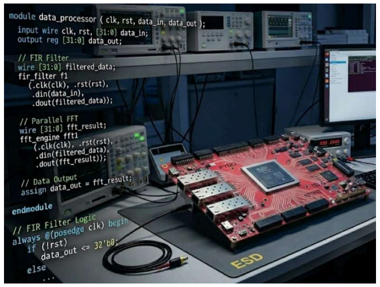
Hardware-Based Real-Time Occupancy Monitoring
A dedicated FPGA implementation eliminates software delays and ensures deterministic, real-time performance.

Bidirectional Detection Using Dual IR Sensors
Entry and exit are accurately identified by analyzing the sequence of sensor activation at the doorway.

FSM-Based Control for Reliable Counting
A Finite State Machine processes sensor signals to avoid false triggering and ensure correct direction detection.

Parallel Processing
Multiple operations (detection, analysis, counting) occur simultaneously, enabling instant response without latency.

Energy-Efficient and Privacy-Preserving Design
Uses IR sensing instead of cameras → low power consumption and no personal data capture.



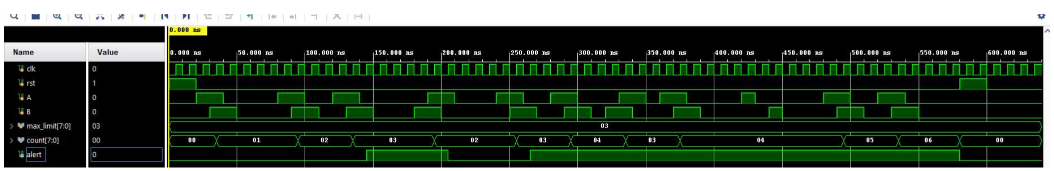
SPECIFICATION & BUDGET

S.No	HARDWARE COMPONENTS	SPECIFICATIONS	COST
1	ASIC Development Budget - Per Chip Cost (Mass Production)	ASIC implementation enables large-scale, low-power, cost-efficient deployment.	Large scale < ₹100
2	IR Break-Beam Sensors (2)	Infrared transmitter-receiver pair, 3.3V/5V compatible	50
6	FPGA Board (PYNQ-Z2) - Only for Simulation purpose	Zynq-7000 SoC, Programmable Logic + ARM Cortex-A9, supports Verilog/VHDL	(Not included)
	Total		**₹200(approx)**

TECHNOLOGY USED

S.NO	SOFTWARE	Purpose
1	Verilog HDL	Hardware design and logic implementation
2	Xilinx Vivado	Design synthesis, simulation, and FPGA programming
3	PYNQ Framework (Python)	Interface with FPGA and testing

COMPLETED WORKS UPTO THIS REVIEW



- **FSM Entry/Exit Sequencing is Accurate:**
  Every A→B sequence correctly increments count by 1 (entry) and every B→A sequence correctly decrements count by 1 (exit)
  Proven clean transitions 00→01→02→03→02→03→04→03→04→05→06 with no missed or double-counted events
- **Alert Triggers:**
  Precisely at max_limit = 3The alert signal goes HIGH exactly when count reaches 03 and immediately drops LOW when count falls back to 02
  Confirms the capacity threshold comparator works cycle-accurately with zero glitches or false triggers
- **Underflow Protection Holds at Zero:**
  After reset brings count to 00, the system never drops below 00 even when exit sequences are attempted
  Proves the underflow guard (count > 0 check) prevents any wrap-around to 255
- **Synchronous Reset is Clean and Instant:**
  When rst = 1 is applied, count snaps to 00 and alert clears to 0 within one clock edge
  FSM resumes normal operation immediately after rst = 0, confirming robust synchronous reset behavior throughout the design

Action Plan

S.No	Major activities (or mile stones) to be completed	Target period of completion	Actual date of completion
1	• Problem definition and requirement analysis • Literature survey and system study • Finalization of project architecture	February 2025	February 2025
2	• RTL design using Verilog HDL • FSM design for direction detection • Functional simulation and verification	March 2025	
3	• Implementation on FPGA (PYNQ-Z2) • Testing with IR sensors and debugging	April 2025	

CONCLUSION

- The proposed system provides an efficient and reliable solution for real-time room occupancy monitoring.
- It uses a hardware-based design (Verilog HDL) instead of software-based approaches.
- FPGA validation ensures correct real-time operation before ASIC implementation.
- Overcomes limitations of existing systems such as: Sequential delays, Timing uncertainties
- Reduced accuracy during continuous movement
- The Finite State Machine (FSM) enables accurate bidirectional entry and exit detection.
- Hardware-level processing ensures deterministic and low-latency performance.
- Supports capacity monitoring and energy management for smart building applications.
- The modular architecture improves flexibility and ease of design.
- The one chip per room approach enhances scalability, reliability, and fault isolation.
- The design is ASIC-oriented, making it suitable for large-scale deployment.
- Provides a low-power and privacy-preserving solution (no camera-based tracking).

REFERENCES

- [1] U. Salmaz, M. A. H. Ahsan, and T. Islam, “High Precision Capacitive Sensors for Intravenous Fluid Monitoring in Hospitals,”IEEE Transactions on Instrumentation and Measurement, 2021.
- [2] M. S. Hossain and G. Muhammad, “Cloud-Assisted Industrial Internet of Things (IIoT) – Enabled Framework for Health Monitoring,”IEEE Internet of Things Journal, 2020.
- [3] Z. Zhang and Y. Wang, “Vision-Based People Counting Using Deep Learning,”IEEE Transactions on Intelligent Systems, 2021.
- [4] A. Kumar and R. Singh, “Arduino-Based Smart Room Occupancy Detection System,”International Journal of Engineering Research & Technology (IJERT), 2019.
- [5] R. S. Gaonkar,Microprocessor Architecture, Programming, and Applications with 8051, McGraw-Hill Education, 2017.
- [6] J. A. Stankovic and I. Lee,“Smart Building Systems for Energy Efficiency and Occupancy Monitoring,”Smart Cities Journal, 2022.

THANK YOU